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AN ECOLOGICAL HISTORY OF INDIA



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PROLOGUE



Prudence and Profligacy

Contemporary India is a fantastic mosaic of fishing boats and trawlers, of cowherds and milk-processing plants, of paddy fields and rubber estates, of village blacksmiths and steel mills, of handlooms and nuclear reactors. Its 850 million people live in tiny fishing hamlets and camps of nomadic entertainers; in long-settled villages and slowly-decaying old towns; in suburban ghettoes and burgeoning metropolitan cities. Some build their shelters with bamboo and mud, others with cement and steel. Some cook with small twigs on a three-stone hearth, or with coconut husks on a mudstove, some with electricity and gas in modern kitchens.

Naturally, the demands of this remarkable mosaic on the country's resources are exceedingly varied. Thus, bamboo is coveted by rural artisans for weaving baskets and to fashion seed drills, by graziers to feed their cattle, and by industry to convert into paper and polyfibre. Landed peasants want the less fertile lands around villages to graze their cattle, the landless to scratch it to produce some grain, the forest departments to produce marketable timber. Peasants want the mountain valleys to grow paddy, power corporations to construct hydroelectric dams.

The resources so varyingly in demand are regulated in equally varying ways. There are sacred ponds which are not fished at all, and beach seiners recognize customary territorial rights of different fishing villages—even though none of this has any formal legal status. Many tribes in north-eastern India own land communally and put it to shifting cultivation, an ownership pattern recognized by law. Village common lands, used as grazing grounds, and wood-lots were once controlled by village communities; they are now government land under the control of revenue or forest departments. There are tenants who cultivate lands that belong to absentee landowners, though much of the land under cultivation is now owned by the tiller. A good bit of the land cultivated by tribals is, however, legally state-owned; this is part of a vast government estate which covers over a quarter of the country's land surface, mostly designated 'Reserved Forest'. Business corporations control large tracts of land, as tea or rubber estates, and there are moves to permit even larger holdings for forest-based industries.

A whole range of these resources, regulated and utilized in many different ways, is under great stress. There are very few deer and antelope left to hunt for hunter-gatherers such as the Phasepardhis of Maharashtra. A majority of the shepherds in peninsular India have given up keeping sheep for want of pastures to graze them. The shifting cultivators of north-eastern India have drastically shortened their fallow periods from a traditional fifteen to a current five years. All over, peasants have been forced to burn dung in their hearths for want of fuelwood, while there is insufficient manure in fields. Ground-water levels are rapidly going down as commercial farmers sink deeper and deeper bore-wells. There are long shutdowns in industry for want of power and raw material, and every urban centre is groaning under acute shortages of housing, fuel, water, power and transport.

In this ancient land, which harbours what is undoubtedly the most heterogeneous of cultures on earth, these resource shortages have given rise to an amazing range of adjustments,

collusions and conflicts. However, the country is living on borrowed time. It is eating, at an accelerating rate, into the capital stock of its renewable resources of soil, water, plant and animal life. Does this mean we are headed for disaster? Or is this a temporary phase before we get back on the path of sustainable resource use? Or perhaps before further technological advances open up an undreamt range of resources? It is obviously true that not *all* resources are being decimated everywhere. In the village of Gopeshwar in the Garhwal Himalaya, for example, there is a nice grove of oak and other broad-leaved species. All families in the village carefully observe traditional regulations on the quantum of plant biomass removed from this grove. But this quantum is quite inadequate for their needs, and for the balance they turn to other hill slopes where exploitation is unregulated and denudation at an advanced stage. This was apparently not the case some decades ago, when the grove was larger and fulfilled the demands of a smaller population in a sustainable fashion.

Human history is, as a whole, precisely such a patchwork of prudence and profligacy, of sustainable and exhaustive resource use. In contemporary India the instances of profligacy clearly outnumber (and outweigh) those of prudence, although this book will argue that such was not always the case. In our own times, acute resource shortages have given rise to a host of social conflicts, and these have significant consequences for what is now happening to the life of India's people and to the health of its land.

The implications of the uses and misuses of India's biological resources are only dimly perceived by the rich and the powerful. But we, in common with the majority of Indians who face the burden of this misuse in their daily lives, believe them to be of tremendous significance: hence this book. We have attempted an ecological history of changing human interactions with living resources, using the Indian case illustratively to explore four themes. The significance of these, in our view, is scarcely restricted to one country, nor even to one continent.

First we ask: under what conditions may we expect human beings to exercise prudence in their use of natural resources?

Second, we investigate both the 'hardware' and 'software' of natural resource use in different historical periods. 'Hardware' refers to the forces and relations of production—namely the technological infrastructure and the systems of property—open access, family, communal, corporate, or state—governing resource use. 'Software' refers to the belief systems (for example, religion, tradition, or science) which legitimize and validate human interactions with nature.

Third, this book analyses the forms of social conflict between different groups of resource users. Here we are especially interested in changes in the intensity of social conflict over time, and in its escalation as one mode of resource use gains ascendancy over another.

Finally, we are interested in the impact of changing patterns of resource use, as well as of social conflict, on the status of living resources. Beginning with the conditions that favour prudence, we come full circle with our analysis of the conditions under which profligacy predominates.

These four themes—i.e. prudence, profligacy, strategies of resource use, and the conflicts to which they give rise—provide the unifying framework for the three parts of this book. The theoretical framework of the study is outlined in Part I. Here we analyse the different forms of restraint on resource use reported from human societies. We locate these practices of resource use in specific cultural and historical contexts. This link between ecology and history is accomplished by means of our typology, which we designate 'modes of resource use'. We put this forward as a supplement to the 'modes of production' scheme which social scientists have traditionally used as a framework for historical periodization.

Part II presents a new interpretation of how the cultural and ecological mosaic of Indian society came together. Given the fragmentary nature of historical evidence on this, our reconstruction may be viewed as *one* plausible scenario of the ebb and

flow of different social systems, as well as of different belief systems of resource use, which existed in the Indian subcontinent before British colonialism. Based on our fieldwork in peninsular India, we advance an ecological interpretation of the caste system, thereby complementing standard economic and ideological interpretations of the persistence of caste as the organizing principle of Indian society.

Part III draws on more abundant source material. It presents a socio-ecological analysis of the new modes of resource use which were introduced by the British, and which have continued to operate, with modifications, after independence in 1947. We argue that British colonial rule marks a crucial watershed in the ecological history of India. The country's encounter with a technologically advanced and dynamic culture gave rise to profound dislocations at various levels of Indian society. While sharply critical of colonialism, Indian historians have, in the main, been indifferent to the ecological consequences of British intervention. Our analysis, in contrast, highlights the essential interdependence of ecological and social changes that came in the wake of colonial rule. Furthermore, our study shows that the socio-ecological consequences of European colonialism in India, while significant in themselves, were quite different from those in the New World.

The British did not merely change Indian history, they also changed the writing of it—by providing historians with an unprecedented level of documentation. Consequently, Part III draws upon a far wider range of sources than the earlier two parts; even so, its focus is only on the most important aspect of the ecological encounter between Britain and India—changes in the ownership and management systems of India's forests. For over a century, India has had the benefits—such as these are—of an extensive system of state forestry. With over one-fifth of the country's land area in its charge, the forest department has been, since the late nineteenth century, by far our biggest landlord. Since students of modern Indian history have shown a surprising unawareness of this fact (cf. Sarkar 1983;

Kumar 1983), and also since every member of India's agrarian society had a direct economic relation to forest produce, a historical assessment of colonial forest management is long overdue. The edifice of colonial forestry has been taken over by the government of independent India. Therefore our analysis will shed light on the links between economic development and ecological change in one important Third World country. Finally, the Indian case, as presented in our history, may illuminate several current debates on tropical deforestation.

In this manner, Part I presents a theory of ecological history; Part II provides a fresh interpretive history of pre-modern India; and Part III contains a socio-ecological history of the forest in modern India.

Although our focus is specifically on interactions between humans and living resources, the business of covering the broad sweep of Indian history is daunting. So we have no illusions about the 'definitiveness' of this enterprise. Our generalizations and data are bound to be modified, altered, even overthrown, in the course of time, as is normal with academic ventures in relatively uncharted terrain. Yet it is undeniable that to date historians of India have been almost completely unaware of the ecological dimensions of social life. Their focus has been more or less exclusively on relations around land and within the workplace, never on the ecological fabric within which both field and factory are embedded, and which these in turn transform. Hence the questions which scaffold this book—the hardware and software of natural resource use, social conflicts around nature, and the cumulative impact on ecological health—are, we believe, asked and partially answered for perhaps the first time in relation to ancient, medieval and modern India.

Ultimately, the ecological history of India must be constructed around detailed regional studies, sharply bounded in time and space. Yet there are periods in the development of scholarship when a new interpretation cannot endlessly await the steady accumulation of certified data. Indeed, to plot the

pieces of a jigsaw puzzle one must begin determining the shape and structure of the puzzle. In the circumstances, while this book provides new data and new interpretations of old data, it provides above all a new and alternative framework for understanding Indian society and history.

Here we draw inspiration from Marc Bloch, who prefaced his great study of French agriculture—itself a model of ecological analysis—with these words:

> There are moments in the development of a subject when a synthesis, however premature it may appear, can contribute more than a host of analytical studies; in other words, there are times when for once the formulations of problems is more urgent than their solution . . . I could liken myself to an explorer making a rapid survey of the horizon before plunging into thickets from which the wider view is no longer possible. The gaps in my account are naturally enormous. I have done my best not to conceal any deficiencies, whether in the state of our knowledge in general or in my own documentation, which is based partly on first-hand research but to a much greater extent on soundings taken at random . . . When the time comes for my own work to be superseded by studies of deeper penetration, I shall feel well rewarded if confrontation with my false conjectures has made history learn the truth about herself.

CHAPTER ONE



Habitats in Human History

Modes of Production and Modes of Resource Use

Many social scientists have found the Marxist concept of modes of production useful when classifying societies according to their technologies and relations of production. Undoubtedly the original scheme, of primitive communism–slavery–feudalism–capitalism, derived largely from the European experience, has been modified by an increasing sophistication within the writing of the histories of non-western cultures. Notwithstanding the problems in applying European models of feudalism to societies such as India and China, as well as the continuing debates around the so-called Asiatic mode of production, the framework itself remains very much in favour. It is strongest while delineating the features of the capitalist mode of production. With the emergence of capitalism as a 'world system', it is currently enjoying a revival in those far-flung corners of the globe where the clash between pre-capitalist relations of production and the capitalist ethos is only now gathering momentum.

Among several important criticisms made of the mode of production scheme, we single out three. The first, made by

Marxists themselves, relates to the relative lack of emphasis in this scheme on political structures and struggles. In his widely-noticed interventions in the 'transition' debate, Robert Brenner argued that the form and intensity of political conflict, rather than changes in production technology or expansion in trade, better explain the nature of the transition from feudalism to capitalism in different parts of Europe (Brenner 1976 and 1978). Other scholars have suggested a supplementary concept, 'mode of power', to more accurately capture the structure of power and domination in different societies (Chatterjee 1983). Second, there are criticisms which, while accepting the relevance of the scheme to European history, express reservations about its application elsewhere. The European model of feudalism does not, for example, fit the Indian experience, and the Asiatic mode is scarcely of any use either, since the state played by no means as important a part in providing public works and irrigation facilities for agriculture as this notion suggests. Finally, there are the criticisms of non-Marxists (and non-economists). These amount to the view that, whatever the merits of the mode of production concept while explaining differences in economic structure, this concept is of little use when interpreting differences in the religious, cultural and ideological attributes of different societies.

While all these criticisms are compelling, from the perspective of this book they do not go far enough. An ecological approach to such questions suggests that the mode of production concept is not adequately materialistic in the first place. This may seem an ironic accusation against a doctrine as supposedly materialist as Marxism, yet a little reflection bears it out. Marxist analyses usually begin with the economic 'infrastructure'—the so-called relations of production and productive forces—without investigating the ecological context, i.e. the soil, water, animal, mineral and vegetative bases of society in which the infrastructure is embedded. As exemplified by recent political and economic histories of modern India, both Marxist and non-Marxist, the most major lacuna in existing

scholarship is an inadequate appreciation of the *ecological* infrastructure of human society. We therefore propose to complement the concept of modes of production with the concept of modes of resource use.

While focusing on spheres of production, such as the field and the factory, most analyses of modes of production have ignored the natural contexts in which the field and factory are embedded—the contexts to which they respond, and which they in turn transform. The concept of modes of resource use extends the realm of production to include flora, fauna, water and minerals. It asks very similar questions. With respect to relations of production, for example, it investigates the forms of property, management and control, and of allocation and distribution, which govern the utilization of natural resources in different societies and historical periods. And with respect to productive forces, it analyses the varying technologies of resource exploitation, conversion and transportation that characterize different social orders.

While complementing the mode of production framework, the mode of resource use scheme incorporates two additional dimensions. First, it examines whether one can identify characteristic ideologies that govern different modes. More importantly, it identifies the ecological impact of various modes, and assesses the consequences of these different modes for the pattern, distribution and availability of natural resources.

Three caveats are in order here. First, the mode of resource use concept, like the mode of production concept, is at bottom an 'ideal type'. Hence, the identification of distinct modes does not preclude the existence of more than one mode in any given social (or, more accurately, socio-ecological) formation. Still, it is usually possible to identify the *dominant* mode within a socio-ecological formation. Second, our treatment is largely restricted to human uses of *living* resources—i.e., flora and fauna—both husbanded and in their natural state. This framework can of course be extended to incorporate other natural resources, such as water and minerals. Finally, one important

respect in which our scheme differs from the Marxian mode of production scheme is that the industrial mode of resource use, as defined by us, includes both capitalist and socialist societies. While there are significant differences between socialist and capitalist paths of development—for example with respect to property and the role of the market—from an ecological point of view the similarities in these two developmental paths are more significant than the differences. For instance, there are structural similarities in the scale and direction of natural resource flows, the technologies of resource exploitation, the patterns of energy use, the ideologies of human–nature interaction, the specific resource-management practices, and, ultimately, the cumulative impact of all these on the living environment in capitalist and socialist societies. Consequently, it makes sense to treat industrial socialism and industrial capitalism as being, ecologically speaking, simply two variants of one industrial mode of resource use.

Four Historical Modes

From the long sweep of human history we can distil four distinct modes of resource use: gathering (including shifting cultivation); nomadic pastoralism; settled cultivation; industry. Later we shall examine the distinctive characteristics of each mode across different axes. These shall include:

- aspects of *technology*, such as sources of energy, materials used, and the knowledge base relating to resource use
- aspects of *economy*, such as the spatial scale of resource flows and the modes of resource acquisition
- aspects of *social organization*, such as the size of social group, the division of labour, and mechanisms of control over access to resources
- aspects of *ideology*, including broad perceptions of the man–nature relationship, as well as specific practices promoting resource conservation or destruction
- the nature of the *ecological impact* itself

After presenting the ideal-typical characteristics of different modes of resource use, we outline the characteristic forms of social conflict *between* as well as *within* different modes. The chapter concludes with a recapitulation, while the appendix contains some reflections on the forms of reproductive behaviour typical of different modes.

Gathering

Technology: The largest period of human history has been spent within the gathering mode of resource use, during which the hunting of wild animals and the gathering of vegetable matter were the mainstay of subsistence. Gathering continues to be significant during the phase of shifting cultivation as well (cf. Elwin 1939; von Fürer Haimendorf 1943a), and we may also include societies that practise shifting agriculture under this rubric. In the gathering mode, societies depend almost exclusively on human muscle power and wood fuel as sources of energy, and on naturally available plants, animals and stones to fulfil their material requirements. Their knowledge base is fairly limited, and nature is viewed as almost totally capricious, as something not subject to human control. The ability to store food and other materials is also very limited, as is the ability to transport materials over long distances.

The economy within this mode of resource use is based on resources which are acquired within a small area of, at best, a few hundred square kilometres. Only a very small range of resources, such as shells, peacock feathers and flint tools, may be transported over larger distances. The diversity of plant and animal matter consumed from the social group's immediate surroundings is high; yet, given this restrictive spatial scale as well as the limited ability to process resources, the actual variety of resources used *in toto* is small, while the quantities consumed are restricted to subsistence needs. Societies which pursue the gathering mode of resource use are highly susceptible to variations within resource availability through space

and time. They respond to such variations by fine-tuned adaptations to local conditions. In the harsher and more variable environments the people comprising these societies subsist as nomadic bands; in the more productive and stable environments they exist as tribal groups confined to relatively small territories. Such territorial restriction also continues with a swich-over to shifting cultivation.

Social organization: The sizes of social groups among hunter gatherer–shifting cultivators are small: kin groups of the order of a few hundred perhaps, largely in face-to-face communication with each other. There are hardly any transactions outside such social groups; the relationship with aliens is largely one of conflict, often over territorial control (Rappaport 1984). The division of labour within these groups is minimal; what exists is primarily based on age and sex, and to some extent upon knowledge and leadership abilities. The women will principally be found involved in gathering plant foods and small animals, while the men will be found hunting the larger animals. As regards the gender equation, men play a greater role in organizing information and taking decisions relating to resource use on behalf of the group as a whole.

In the gathering mode there is little variation among members of a group in terms of access to resources, and notions of private property are extremely poorly developed. Within a group, here, no individual is in a position to dominate and coerce others to any significant degree. While within-group differences thus tend to be low, the differences between groups may or may not be equally low. Till 10,000 years ago, before plants and animals began to be husbanded by humans, between-group differences were, without doubt, at a low level. The human populations of any region might then have been divided amongst a large number of endogamous groups competing with each other for control over land and water. This competition would have been intense in productive and stable environments, such as tropical rain forests, where inter-tribal warfare over territorial rights was probably a routine occur-

rence. On the other hand, the limits of territories are likely to have been much fuzzier in harsher, highly variable environments, such as tropical deserts or the coniferous forests of higher latitudes. Here, inter-tribal conflicts would have been correspondingly less acute. With the beginnings of cultivation and animal husbandry, and later the industrial revolution, people who pursued the gathering mode came to be increasingly disadvantaged *vis-à-vis* people with access to advanced levels of technology.

Economy: We may characterize the economy of gatherers as a *natural* economy, in so far as it draws all its resources directly from nature (Dasmann 1988, Worster 1988). Here, the flows of resources are largely closed on spatial scales of a few hundred, or at best a few thousand, square kilometres, over which each endogamous social group of gatherers might range (Fig. 1.1). While there may be some flows of materials, such as shells or stone tools, over longer distances, these are insignificant (in terms of physical mass) when compared to food and the other resources utilized by the human groups concerned. The material flows would, therefore, not only be closed, they would also be balanced, i.e. with inflows matching outflows on a fairly restricted spatial scale. This balance is apt to be disrupted when gatherers come into contact with people practising more advanced modes of resource use. The latter are then likely to organize net resource outflows from regions inhabited by gatherers, and perhaps also affect such regions through inflows of wastes—of materials they discard.

Gatherer societies, as indeed all human societies, encounter three distinct situations with respect to their resource base. First, their demands on the resource base may be small compared to the overall availability. This may be due to their having colonized a new habitat—as must have happened when some groups first crossed over into North America—or on account of a technological innovation such as the bow and arrow, which opened up a new range of species to prey upon. This is a situation analogous to that of r-strategists in ecological litera-

ture, i.e. of populations that increase relatively rapidly for most of the time. The r-strategists include weedy species which rapidly colonize disturbed environments, or micro-organisms like influenza which spread in epidemics. Other societies may be in equilibrium with their resource base, as for instance fishing communities in an isolated coral island. These are analogous to the k-strategists of ecological literature, examples of which include shade-tolerant and timber-tree species in rain forests, or micro-organisms like tubercular bacilli. Finally, societies may be confronted with a shrinking resource base, as may have confronted gatherer societies who were displaced by advancing glaciers, or by technologically more advanced human societies. Ecologists have no specific term for this situation. However, given the fact that some twenty times as many biological species have gone extinct as are alive today, many biological populations must be adapted to dealing with a shrinking resource base.

Under conditions approaching equilibrium with their resource base, gatherers exist in kinship-based, small, viscous groups that are more or less tied to specific localities, depending on the variability of the environment. Such an organization might be broken up when gatherers come in contact with people who work within more advanced modes of resource use, and consequently have to adjust to a shrinking resource base. In the former case, the group's members would tend to be involved in a network of co-operative behaviour, and would be very sensitive to group interests. In the latter case, such co-operative behaviour, as also the perception of group interests, may dissolve.

Ideology: To gatherers, with their limited knowledge base, nature follows its own capricious ways, hardly subject to human control. Gatherers typically regard humans as merely part of a community of beings that includes other living creatures, as well as elements of the landscape such as streams and rocks. Especially where gatherers are attached to particular localities, as in productive and stable environments like tropical humid

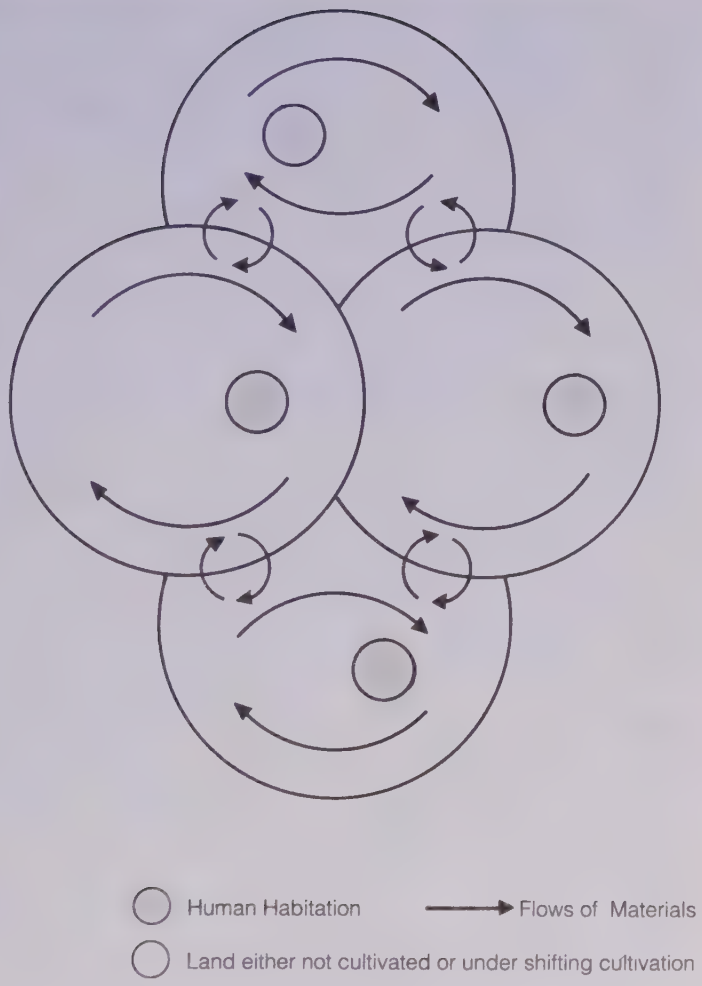


Figure 1.1
In productive, stable environments, hunter-gatherer cum shifting-cultivators maintain well defined territories. Cycles of materials in such environments are largely closed on the spatial scales of territories with flows of materials across territorial boundaries being much less significant. The thickness of an arrow indicates the intensity of the flow.

forests, they attribute sacred qualities to individual trees, ponds or mountain peaks, or to all members of a plant or animal species, such as *Ficus* trees. They often treat plants, animals or elements of the landscape as kin, or as being in relationships of either mutualism or antagonism. Thus, rivers may be considered as mothers, and totemic animals (like bears or antelopes) as brethren; specific trees may be seen as inhabited by demons who need to be placated. Gatherers therefore enter into a whole range of frequently positive relationships with these other 'beings' of their own locality. By the same token, they have no relationship with plants, animals or landscape elements outside their own locality (Martin 1978; Macleod 1936).

Restraint—Real and Apparent: At a more concrete level, these ideologies of nature worship are buttressed by *specific social practices* which orient societies in the gathering mode towards the prudent use of nature. Thus, many gatherer–shifting cultivator societies have a variety of practices which regulate their behaviour towards other members of their community of beings, and which seemingly contribute towards ensuring the long-term sustainability of resource use (Gadgil 1991; Gadgil and Berkes 1991; McNeely and Pitt 1985; Ruddle and Johannes 1985). In the context of ecological debates on prudence and profligacy, it is of interest to examine these practices in order to assess whether they could be better explained in terms of harvesting for short-term gain; and if not, whether they could indeed lead to an enhanced availability of resources to the group as a whole in the long run. These practices, studied by anthropologists and ethnobiologists, involve a variety of restraints on harvesting in terms of quantity, locality, season, and life-history stages. They also involve differential harvests by age, sex, or social class. It is, of course, possible that such apparent restraint may have nothing to do with a long-term conservation of the resource base. A harvester interested in calculating an immediate return may still not use a resource if the net gain obtainable from it is below a certain threshold, which in turn would depend on the net gain obtainable from

alternative resources. We must therefore examine each supposed instance of restraint to assess whether it could involve such a discontinuation of resource use. This may happen, for instance, because the cost of harvest increases excessively (Smith 1983; Borgerhoff Mulder 1988).

The whole range of practices demonstrating the restrained use of resources by human beings can be classified under ten broad categories:

- A quantitative restriction on the amount harvested of a given species, or from a given locality, by harvesters. The imposition of such quotas implies that the harvest is halted at resource densities greater than those at which individuals would find the net gains too low to continue harvesting. As a corollary, these quotas are likely to enhance total yields over the long-term, and at the sacrifice of some immediate return. These are therefore likely to be genuine instances of restraint.
- Harvesting a certain resource may be abandoned when resource densities fall. In parts of New Guinea, for example, the hunting of Birds of Paradise is temporarily abandoned if their populations decline (Eaton 1985). This sort of response is expected from harvesters who are attempting to maximize a short-term net gain, since a fall in the resource density would progressively increase the costs of harvesting. It is possible, though not very probable, that harvesting is abandoned well before this level is reached in the interest of long-term yields.
- Harvesting from a certain habitat patch may be abandoned if yields from that patch are reduced. Thus, in the Torres Strait, fishing may be stopped in regions where fish yields are known to have declined (Nietschmann 1985). This again is a response which is to be expected from a forager who is attempting to maximize immediate net returns. It can be related to long-term resource conservation only if concrete quantitative evidence is available to indicate that harvesting is abandoned in advance of the returns reaching a value low enough to justify abandoning harvests.
- Harvesting from a certain species may be abandoned in a

certain season. Illustrative here is the taboo, in many Indian villages, on hunting certain animals between July and October (Gadgil 1985a). Possibly this taboo is a consequence of returns which are too low to justify harvesting for immediate gains in that season. Conversely, if in fact net returns in that season are expected to be relatively high, this is likely to be a conservation measure.

• Harvesting from a certain habitat patch may be abandoned in a certain season. Again, this could be a response to an excessively low level of net gain from that habitat patch in that season. This should be verified by a comparison with net gains in other seasons, and if possible by a quantitative assessment.

• Certain stages of life—by age, sex, size or reproductive status —may immunize an animal or bird from being harvested. For example, in the village of Kokre-Bellur in the state of Karnataka, birds in the breeding within a heronary may be left unmolested, though they may be hunted elsewhere and at other seasons (Gadgil 1985b). If such protected stages appear to be critical to population replenishment, and if they are likely to yield as high or higher net returns than the unprotected stages, it is reasonable to assume that this measure is designed specifically to conserve the resource. On the other hand, if these stages are likely to yield lower net returns in comparison with the unprotected stages, then they might be left unharvested simply in the interest of maximizing the immediate net gain.

• Certain species may never be harvested either because of the relative difficulty of procuring them (risk of injury during the hunt) or because they carry parasites that can affect humans. If these conditions do not operate, conservation can indeed serve the long-term interest of human resource use: if the species thus protected enhances the availability of some other species that are harvested. This is likely for a widely protected species, such as trees belonging to the genus *Ficus*, but much less likely for a wide variety of species protected as totemic by certain tribes (Gadgil 1989).

• Certain habitat patches may either never be harvested, or be

subject to very low levels of harvest through strict regulation. It is extremely difficult to arrive at workable prescriptions on quantitative quotas, closed seasons, or protected life-history stages that decisively guard against resource decimation. Providing refugia (sacred groves, sacred ponds, etc.) may then be the most easily perceived and most efficient way of guarding against resource depletion (Gadgil and Vartak 1976).

- Certain methods of resource harvest may be wholly prohibited or strictly regulated. Thus, fishing by poisoning river pools is severely regulated by tradition in many parts of India (Gadgil 1985a). If these methods are likely to provide as high, or higher, net returns than the permitted methods, their regulation may serve the interest of long-term resource conservation.
- Certain age/sex categories or social groups may be banned from employing certain sorts of harvesting methods, or from utilizing certain species or habitat patches. Thus in New Guinea adult males are banned from hunting rodents (Rappaport 1984). This could contribute towards long-term resource conservation by moderating the total amount of harvest. It could also assist in long-term conservation by restricting the access to a limited number of individuals who may more readily come to use the resource in a prudent fashion. It is, of course, quite possible that such restrictions merely benefit segments of the community that are in positions of power, without really serving the interest of long-term conservation.

Simple Rules of Thumb

Given the complexity of ecological communities, precise prescriptions for the prudent use of living resources are difficult. Detailed quantitative prescriptions seem impossible, given the present state of knowledge (Clark 1985). This is particularly so if the entire prey population is continually subject to being harvested. But certain simple prescriptions to avert a resource collapse seem easy to formulate and should have a significant effect in enabling sustainable use. These prescriptions are five-

fold: provide complete protection to certain habitat patches which represent different ecosystems, so that resource populations are always maintained above some threshold; provide complete protection to certain selected species so that community-level interactions are disrupted minimally; protect such life-history stages as appear critical to the maintenance of the resource population; provide complete protection to resource populations at certain times of the year; and organize resource use in such a way that only relatively small groups of people control or have access to a particular resource population (Johannes 1978).

Modern ecological and evolutionary theory suggests that such prescriptions are likely to assist in avoiding an environmental collapse, although they would by no means ensure harvests at maximum sustainable yield levels. In his classic experiments on prey-predator cycles of protozoans, Gause showed that prey extinction could be effectively avoided only by providing the prey a refugium, an area of the experimental arena inaccessible to the predator where the prey could maintain a minimal population and from which other areas could be colonized by it (Gause 1969). Sacred groves, sacred ponds, and stretches of sea coast from which all fishing is prohibited are examples of such refugia. Modern ecological theory also stresses the significance of certain species which serve as keystone resources or mobile links in maintaining the overall functioning of the community (Terborgh 1986). The tree of the genus *Ficus*—to which belong species such as banyan and peepal (widely protected in Asia and Africa)—is one such keystone resource. Contemporary ecological theory also points to the fact that certain stages in a population are of higher 'reproductive value', and therefore more significant in permitting continued population growth. Pregnant does and nesting birds, again often protected by humans, are such stages (Fisher 1958; Slobodkin, 1968). Finally, recent work on the evolution of co-operative behaviour emphasizes that restraint is progressively more likely to evolve if the number of individuals involved in repeated

social interactions gets progressively smaller (Joshi 1987; Feldman and Thomas 1986; Berkes and Kence 1987).

It thus appears plausible that, over the course of human history, there have been human groups whose interests were strongly linked to the prudent use of their resource base, and that such groups did indeed evolve conservation practices. Many of the practices described above have in fact been reported from different gatherer societies. These conservation practices were apparently based on several simple rules of thumb that tended to ensure the long-term sustenance of a resource base. These rules of thumb were necessarily approximate and would have been arrived at by trial and error. Practices which seemed to keep the resource base secure may have gradually become stronger; conversely, there could have been a gradual rejection of those practices which appeared to destroy the resource base (Joshi and Gadgil 1991). Conservation practices observed by various other social groups may also be emulated if they appear successful in resource conservation. Such a process is likely to lead to the persistence of a whole range of practices, some beneficial from the point of view of resource conservation, but also others that are neutral; and perhaps some that might once have been beneficial or neutral but later prove harmful on account of changed circumstances.

Diversity of resources: In conjunction with their various practices of restrained use, gatherer societies are remarkable for the great diversity of biological resources they utilize. Studies on the American Indians of Amazonia have shown that they utilize several hundred different species of plants and animals for food, and as sources of structural materials and drugs. They have distinct names for as many as 500 to 800 biological species (Berlin 1973). For another thing, different tribal groups may be familiar with and utilize a different set of species. As early as 30,000 years ago, for example, two Neanderthal groups of Dordogne in France had apparently specialized in different prey species, one group concentrating on horses and the other on reindeer (Leakey 1981). We may therefore conclude that primi-

tive gatherer-shifting-cultivator societies valued a very wide range of biological diversity, and evolved cultural practices which promoted the persistence of this diversity over long intervals.

Deliberate destruction: While gatherer societies typically have a variety of practices that could help conserve the resource base of their own localities, they may also deliberately destroy the resource bases of aliens. Thus, when New Guinea highlanders defeat and drive away a neighbouring group from its territory, the conquerors do not immediately occupy this territory. They cut valued fruit-yielding trees from the conquered group's territory, thereby rendering it far less desirable for recolonization by the conquered. The actual territory may be physically occupied only later, if it is not reoccupied by the vanquished group (Rappaport 1984).

Ecological impact: Gatherer societies, with their low population densities, low per capita resource demands, cycles of materials closed on limited spatial scales, and a number of practices that promote sustainable resource use, necessarily have a low level of impact upon the environment. Over long intervals, however, even this can add up to substantial changes. Such changes are especially likely when the resource base changes relatively rapidly, as might have happened with fluctuations during the ice age; or when a gatherer population encounters an entirely new resource base, as during the initial colonization of the Americas. It has, for instance, been suggested that the widespread extinction of large mammalian species during the Pleistocene was a consequence of human overhunting, and that many of the savanna-grassland formations of East and South Africa are a result of fires lit over tens of thousands of years by human populations (Menant *et. al* 1985). Nevertheless, the pace of such impact would be considerably slower than the pace in populations which possess more advanced modes of resource use, as described later in this chapter. Some writers have claimed that hunter-gatherers possess an ecological wisdom far in advance of that shown by modern man (Shepard 1982). Be

that as it may, it is indisputable that the ecological impact of this mode of resource use is minimal.

Pastoralism

Technology: The long period of history when human beings were exclusively gatherers began to come to a close with the domestication of plants and animals. This coincided with the withdrawal of glaciers, ten thousand years ago. It is possible that climatic and vegetational change prompted human populations to intensify resource use, and to initiate agriculture and animal husbandry. These processes began in parallel, and have often gone hand-in-hand. While the cultivation of plants has been of greater significance in tracts of moderate-to-high rainfall and moderate-to-high temperatures, animal husbandry has held pride of place in tracts of low rainfall, and at the higher altitudes and latitudes where temperatures are too low to support agriculture (Grigg 1980). Over large tracts where agriculture is not feasible, it is also difficult to maintain herds of domestic animals within a single locality. Animal husbandry is therefore based, in such tracts, on moving herds from place to place often over several hundred kilometres. This requires taking advantage of the seasonal abundance of grazing resources in different parts of a region. Nomadic pastoralism thus evolved as a distinctive mode of resource use, a mode that held sway for several centuries over large regions, particularly in Central Asia and North and Central Africa (Leeds and Vayda 1965; Forde 1963).

Pastorals have access to animal muscle power, an important additional source of energy, especially for transport. The animals also serve as a source of food which can be tapped as required, thus greatly increasing flexibility in the use of different habitats.

Economy: Nomadic pastorals move over large distances, and with their access to animal energy, have been critical in creating flows of resources over distance scales that are vastly greater

than those which prevail in gatherer societies. The resources they moved have been both high-bulk commodities like salt, and high-value, low-bulk luxury items like precious stones and musk. They also served as carriers of information about resources of distant regions and of technologies elaborated by other societies. Consequently, pastorals not only continued some hunting-gathering while on the move, and produced meat, milk, hide and wool from their animals, but also acquired resources, especially from settled agricultural societies, in exchange for material and information. Even more important, nomadic pastorals could effectively deploy force to usurp resources from societies of cultivators, as Chengiz Khan did with great success over the huge regions of Asia and Europe. Indeed, nomadic pastoral societies, in their heyday, may have behaved much like the r-strategists of ecological literature.

Social organization: The social groups of nomadic pastorals remain limited to kin groups of a few thousand; nevertheless, they come in contact with large numbers of other groups over an extensive terrain. Within the social groups of pastorals, the division of labour is fairly limited. It is based on age, sex, and leadership qualities which emerge during inter-group conflict. Women may be more involved in feeding, milking, and tending animals, men in deciding on migration routes and herding animals while on the move.

In the pastoral mode, elements of private property begin to emerge. However, while herds are usually owned by separate households, pastures are invariably common property, with individual herdsmen possessing rights of access and usufruct. However, like gatherer societies, nomadic groups are relatively egalitarian (Khazanov 1984). With this, coercion within groups remains limited; indeed there is considerable premium on co-operation within the group, especially in the context of conflict with other groups, nomadic or otherwise.

Ideology: By surviving successfully in harsh and variable environments, and with little attachment to any particular locality, nomadic pastorals were perhaps the first societies to

perceive human communities as separate from nature, and therefore in a position to dominate it. Since the usurpation of resources controlled by alien and settled communities constituted a significant part of their strategy of resource acquisition, they were unlikely to evolve strong traditions of careful or restrained resource use. Indeed, ideologies which rejected the attribution of sacred value to living creatures or to natural objects, e.g. religions like Judaism, Christianity and Islam, arose in tracts that were dominated by nomadic pastorals in the Middle East. In fact, as Lynn White (1967) notes, such religions sometimes prescribed the deliberate destruction of sacred trees and sacred groves.

The ritual life of nomads is quite meagre: no pantheon of gods, as in peasant societies, no system of totemism, as in gatherer societies. Ritual importance may be placed on livestock, but almost never on natural locations or specific fields. Equally striking is the relative unimportance here of witchcraft. In comparison with peasants and gatherers, nomadic pastorals have little need to pacify or placate nature; in the event of resource shortages they remove themselves to more resource-abundant areas—something peasants cannot do (Goldschmidt 1979).

Apart from these broad ideologies, in their day-to-day interaction with nature nomads do have practices which reveal a deliberate restraint on resource use. These practices include the complete exclusion of grazing pressure during certain periods within fodder reserves, and its limited use during other periods in terms of the kind and number of animals permitted for grazing—as for example in the system of *ahmias* around Taif in Saudi Arabia (Draz 1985).

Ecological impact: It is possible that nomadic pastorals contributed to a gradual overgrazing, and to the expansion of arid regions at their margins, all through their history. This they have certainly done across many regions in modern times. They have also contributed to ecological degradation through the organization of trade and the diffusion of technology over large

distances and perhaps most importantly by disseminating the belief in man's mastery over nature.

Settled Cultivation

Technology: Human societies learnt to cultivate plants and domesticate animals around the same time, beginning some 10,000 years ago. In some regions the two developed hand-in-hand, with the traction power of animals and the manurial value of their dung being vital to agricultural operations. This, for instance, was the case in the Middle East, from where the use of cattle and the plough, and the cultivation of wheat and barley, gradually spread over parts of Asia, Europe and Africa. In other areas, domesticated animals played a much less significant part in cultivation, as in the paddy-growing tracts of Asia—or had no role at all—as in the case of maize cultivation in pre-Columbian America (Grigg 1980).

Cultivation involves an intensified production of certain species of plants, and the removal of plant material from a relatively restricted area of land. The plant material so removed, for instance cereal grains, are particularly rich in certain elements, such as nitrogen and phosphorus, and contain a number of micronutrients, like boron and molybdenum, in smaller quantities. The continuation of cultivation on a piece of land therefore depends on returning to the earth what is taken away from it. This happens either through long periods of fallow, as in shifting cultivation, or by the application of river silt, organic manure, or mineral fertilizers if the same piece of land is tilled year after year. Shifting cultivation is, of course, the option followed so long as the amount of land available is large relative to the population. As this ratio declines, the same piece of land has to be used more and more intensively. Almost everywhere, this has called for the extensive use of organic manure derived from natural vegetation in the surrounding areas, gathered either through grazing domestic animals or directly by human effort. This has changed radically only in

recent times, when fossil fuel energy began to be used to efficiently mine, transport and synthesize mineral fertilizers to augment agricultural production (Pimentel and Pimentel 1979).

Through most of its history, settled cultivation has thus depended on human muscle power, supplemented in some regions by animal muscle power. In the industrialized world, it has come to depend increasingly on fossil-fuel energy. However, pre-industrial agriculture depends primarily on plant and animal-based materials, along with some control of natural flowing water for irrigation. Consequently, pre-industrial agricultural societies (more properly, peasant societies) have a fairly substantial knowledge base in relation to husbanded plants and animals; they also view nature as being subject to human control to a very significant extent.

Economy: In peasant societies, cereal grains can be stored and moved around, especially on the backs of animals or in carts, over long distances. Resources can here flow over much larger distances than in gatherer societies, enabling the concentration in towns of human populations not directly involved in gathering or the production of food. Changes in settlement patterns also correspond to shifts in consumption. Of course, a majority of the agrarian population consumes natural resources largely for subsistence—e.g. for food, clothing, shelter, implements, fodder and manure. However, a small but powerful segment of the population is involved in the large-scale consumption and use of materials not directly related to subsistence—both luxury items such as silk and wine, and instruments of coercion such as the horse and elephant, metal swords and shields. Fig. 1.2 shows the resulting structure of material flows in such a system. There are large-scale exports of materials out of intensively cultivated patches of lands, both to nearby villages and to more distant urban centres. The volume, range of items and distance-scale of such flows steadily increase with technical improvements, especially in animal-based transport. These outflows from agricultural land are balanced by inflows from surrounding non-cultivated lands. In peasant

societies there are no counterflows from urban centres back to cultivated lands. However, as Fig. 1.3 shows, in industrial societies there are large flows, back to the land, of materials such as farm machinery and synthetic fertilizers.

Characterized by fairly extensive resource flows, especially of foodgrains and livestock, peasant societies are much less subject to environmental variation in space and time. Nevertheless, the techniques of cultivation and animal husbandry, the choice of plant and animal varieties, and the way that resources of non-cultivated lands are gathered and put to use are all greatly dependent on the local environment. There is therefore a substantial degree of locality-dependent adaptive variation in patterns of resource use within peasant societies. This begins to disappear only with large-scale inputs of fossil-fuel energy and advanced technology into agriculture.

Social organization: Cultivation requires intensive inputs of human energy in relatively restricted areas of land—a few hectares per person in the pre-industrial stage. Therefore a small kin group rather than a large band can most effectively organize such inputs; hence the family becomes the basic unit of an agricultural society. Family groups need to co-operate with each other in a variety of ways, including defence against the usurpation of their production (apart from what is surrendered as tax to the state). Several families thus remain banded together in a village, which becomes a social group of a few hundred to a few thousand individuals. This social group also tends to control and manage a territory of non-cultivated land surrounding the cultivated areas, from which come a variety of inputs such as fuel-wood, fodder and leaf manure.

Sex-based division of labour is quite pronounced in the peasant mode. Typically, men confine themselves to operations such as ploughing, which require higher power output. Women take on the burden of more tedious work, such as weeding and transplanting, and, outside cultivation, the collection of fuel, fodder and water.

While cultivated plots are usually (though not always) con-

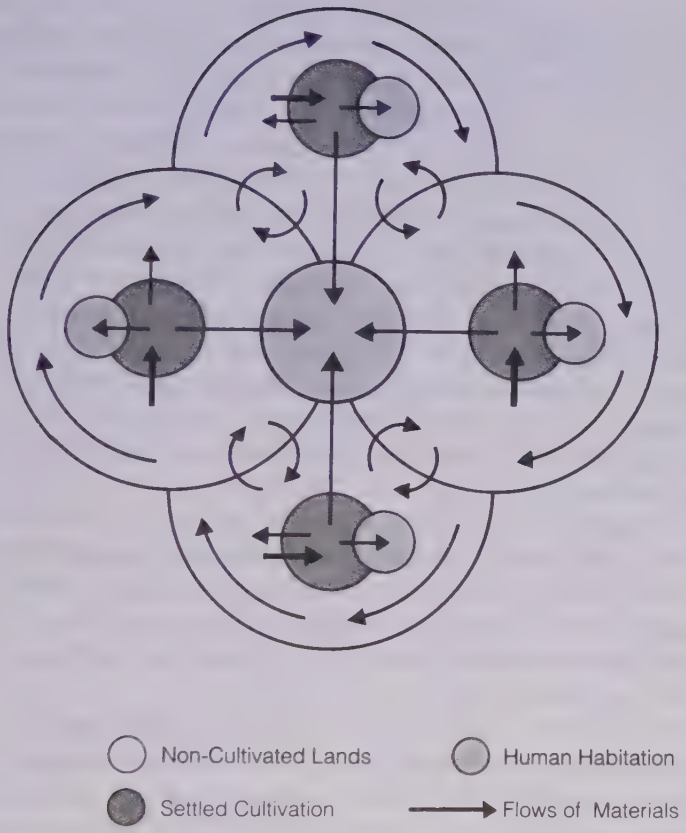


Figure 1.2

Material flows in an agrarian society. Settled agriculture makes possible generation of surplus grain and livestock production which can support concentration of non-agricultural populations in towns and cities. This material export from cultivated lands has to be made good by flows from surrounding non-cultivated lands. Material cycles thus become much more open in comparison with the hunter-gatherer shifting cultivator stage. Settlements adjacent to cultivated land represent villages, the larger habitation in the centre, towns. The thickness of the arrow indicates the intensity of the flow.

trolled by individual households, forests, grazing grounds and water are normally held in common by the village (cf. Mukerjee 1926). Several, and sometimes very many, such villages are integrated into a larger chiefdom, which constitutes the terrain over which the surplus of agricultural production is pooled together. The right to do so is contested with neighbouring states. The larger social entities include concentrations of non-agricultural populations in larger settlements, the towns or cities with thousands or tens of thousands of inhabitants. While the villages constitute a face-to-face social group (not necessarily kinship based) of several hundred or thousand people analogous to the bands/endogamous tribal groups of gatherers, the larger society integrated through larger-scale resource flows can no longer deal with all members of the group in a personalized way.

Within the larger social group of an agricultural society there is a great deal of division of labour, made possible by the fact that only a fraction of the population needs to engage directly in the gathering and production of food. Those not directly involved in the production of food take on other occupations. These are—

The processing of materials (e.g. textiles, oilseeds), transportation, the interpretation and dissemination of natural and cultural knowledge (by priests) and coercion (by specialized warrior groups). In this division of labour, men end up monopolizing the more prestigious and skilled jobs, passing on the less skilled and tedious work to the women. There is now a substantial differentiation of coercive abilities within the social group—from peasants who receive very little in return for the surplus production they yield to others, to priests and warriors who provide others little in return for the surplus they manage to get hold of (Service 1975).

In comparison with the gatherer mode, the peasant mode shows a sharp separation between cultivated and non-cultivated land. This separation is significant in directing resource flows (Fig. 1.2), and equally so with regard to differing forms

of property and control. At the lowest spatial scale agricultural land may be controlled by a family. Such control may be subject to regulation by the village community, which could reassign plots of land and, further, treat land as a community resource, perhaps for grazing purposes, outside the cropping season (Bloch 1978). The non-cultivated land within village boundaries, typically a few square kilometres in extent, serves to supply fuel, grazing, manure, etc. for the community as a whole. These large chunks of land, different portions of which may be used at different seasons, may be most effectively controlled as community, rather than family, property. (This also applies to water sources such as tanks, rivulets, lakes and springs, typically held in common.) However, with technological advance and the concentration of powers of coercion, village communities may lose control over cultivated lands and become tenant-cultivators. Access to non-cultivated lands may be lost too, for instance with the enclosure of commons by powerful landlords or by the state. In addition to land in the vicinity of villages, the state may lay claim to larger uninhabited tracts, constituted as princely hunting preserves or forests from which the army derives its supply of elephants (Thompson 1975; Trautmann 1982).

The large-scale resource flows of agricultural societies are accompanied by imbalances involving net outflows from rural areas and inflows into urban centres. These imbalances become progressively more acute as technological advances in storage, transport and processing create an effective demand for a larger range of produce from both cultivated and non-cultivated lands, and as the coercive power of the non-agricultural sector increases *vis-à-vis* the agricultural sector. The overall imbalances decrease with large-scale inflows of synthetic chemicals, fertilizers and machinery to agricultural lands in industrial societies; however, it is very likely that, given the larger inflows and outflows, the imbalances pertaining to specific elements such as micro-nutrients, or to larger organic molecules like

humic acids, may become all the more acute (Pimentel and Pimentel 1978; Stanhill 1984).

Agricultural societies might be at equilibrium with their resource base; or they might encounter either expanding or shrinking resource bases. Agriculturists who newly colonize lands earlier held by gatherer societies, or who newly benefit from a major resource input such as irrigation, would find themselves with an expanding resource base, analogous to the r-strategists of ecology. On the other hand the base may be shrinking because of an adverse climatic change, or if access is cut off from important inputs like leaf manure and fodder from forests newly taken over by the state. Their resource base may also shrink if agricultural productivity remains stagnant in the face of human population growth. An approximate equilibrium may be maintained if the population grows slowly, if the external demands on agricultural production remain stable, and if technological progress keeps pace with the need to continue increases in agricultural production in consonance with population change.

In the last case—that of approximate equilibrium as with the k-strategists of ecology—the social groups are likely to be highly viscous, with related individuals tending to stay together and tied, perhaps generation after generation, to a given locality. Under these conditions they may exhibit high levels of co-operative behaviour among themselves, as also behaviour which favours long-term group interests. The peasant societies of India, China and South East Asia, in the period before European colonization, perhaps fall into this category. On the other hand, when the resource base is rapidly expanding, especially with new land being brought under cultivation, social groups are likely to be much more fluid and far less tied to any locality. Their level of co-operative behaviour, and especially their willingness to sacrifice individual interests to long-term group interests, would consequently be much lower. This would seem to have been the case with European pioneers in seventeenth and eighteenth century North America. Finally,

peasant cultures faced with a shrinking resource base may also lose group coherence and the attachment to a particular locality, as has been reported from several parts of India in recent times (Guha, 1989a).

Ideology: In comparison with hunter-gatherer societies, agricultural societies have established substantial control over natural processes; nevertheless, they are still very subject to nature's caprices in the form of droughts, floods, frost, and plagues of locusts. Indeed, certain life-history stages of agricultural crops are themselves especially sensitive to the environment. Hence, agricultural societies continue, in part, to perceive man as one among a community of beings. At the same time, the image of man as a steward of natural resources acquires influence. The restrained use of natural resources could thereby be expected to form one part of the ideology of agricultural societies, especially when they are in a state of near equilibrium with their resource base. On the other hand agricultural societies in the process of encountering an expanding resource base—either through new technologies or, especially, while colonizing lands earlier held by gatherers—are much more likely to view man as separate from nature and with a right to exploit resources as he wishes (cf. Cronon 1983).

In stable peasant societies, practices of restrained use relate to cultivation itself, and are linked to a philosophy of minimizing risk rather than maximizing immediate profit (Scott 1976). The use of a whole variety of different crops and crop rotations, and the careful community-based maintenance of irrigation ponds, may be part of such an approach. This approach would also encompass the non-cultivated lands from which the villagers gather fuel, fodder, small timber, leaf manure, and so on. A variety of practices of restrained resource use with respect to non-cultivated lands have been reported from peasant societies which are at equilibrium with their resource base. For example:

• A quantitative restriction on the amount of harvest from a given locality, e.g. on the amount of wood or grass harvested by a family or their livestock from community lands.

• Restrictions on the harvest in certain seasons. Thus, lopping the green leaves of trees may be permitted only after the rainy season, i.e. after the trees have ceased to put on growth.
• Certain species, e.g. trees belonging to the genus *Ficus*, may be wholly protected.
• Certain habitat patches may never be harvested. In the state of Mizoram, in north-eastern India, the community wood-lots from which regulated harvests are permitted, called supply forests, are complemented by sacred groves, aptly called 'safety forests', from which no harvests are permitted.
• Certain methods of harvest may be completely prohibited. In the Aravalli hills of Rajasthan there are patches of forests called *oraons* where harvesting by using metal tools is prohibited, though wood may be removed by breaking twigs by the hand (Brara 1987).
• Specific age-sex classes or social groups may be banned from employing certain harvesting methods, or from utilizing certain species or habitat patches.

These practices all depend on a high degree of co-operation between the members of a village community. One could say that, in the peasant mode, *custom* and *tradition* provide the overarching framework within which human–nature interactions are carried out. While religion continues to permeate social life, in the realm of resource use it is supplemented to a significant degree by custom. In other words, customary and time-honoured networks govern relationships of reciprocity within peasant society—as, for example between different castes in a village (cf. Chapter 3), between nomadic pastorals and peasant cultivators, or between the village and the state. Clearly, such relations (with respect to resource use, as elsewhere) are often asymmetrical: but they normally fluctuate only within the limits defined by custom.

Ecological impact: The ecological impact of the peasant mode may be characterized as *intermediate*. With the march of agriculture a significant proportion of land begins to be converted into artificial grasslands or cropfields, which replace forests, mar-

shes or natural grasslands. Fire, stone axes and metal axes aid in this process of conversion. Cultivation also imposes increasing demands on natural vegetation and a greater removal of forest produce, to be used as fuel, fodder, manure, building-timber and implements. The discovery of iron, which in many areas led to the colonization of the forest by agriculturists, also facilitates the continued felling of individual trees in a forest. In villages where such felling takes place, natural regeneration is relied upon to restore tree cover. At the same time, improvements in weapon technology enable a more flexible hunting strategy. The cumulative impact of these interventions is a striking change in the landscape, which very likely becomes heterogeneous, manifesting a variety of successive stages within a mosaic. It could also result in the local extinction of some species of plants and animals.

Of course, agricultural societies which newly colonize lands held by gatherers have had a dramatic ecological impact even in the short run—transforming the landscape, exterminating certain species and depleting others, introducing weedy species, and so on (Cronon 1983; Crosby 1986). This would also be the case with agricultural societies expanding their resource base through technological innovations such as large-scale irrigation and the use of pesticides. On the other hand, agricultural societies in approximate equilibrium with their environment—dominated by 'local production for local use'—have only moderate levels of impact in transforming landscapes and bringing about gradual changes in the composition of biological communities.

The Industrial Mode

Technology: The latest mode of resource use to appear in human history, large-scale industry, has been with us for just about two hundred years. This is only one-fiftieth of the time that *Homo sapiens* has spent husbanding plants and animals, and one-two hundredths of the time since when hunter-gatherers painted

grand hunting scenes in the caves of Lescaux and Altamira. But its ecological impact has been profound, far surpassing all that preceded this revolution. The main reason for this is the quantum jump in the use of energy, with heavy demands on non-renewable sources (coal, oil), coupled with the use of entirely novel sources such as nuclear energy.

If the pattern of energy use in the gatherer mode may be characterized as *passive* (relying only on human-muscle and wood-fuel power) and that of the agricultural mode as *active* (augmenting human power with animal power, wood-fuel and water power), in the industrial mode energy use follows an *extractive* path, wherein natural resources are both *harnessed* (hydro power) and *mined* (fossil fuels) for human consumption. The industrial mode has also brought into use a whole new range of man-made materials; e.g. metals, plastics, silicon chips and synthetic pesticides. These newly-fashioned materials can now be preserved to be used for long periods, and transported for consumption elsewhere. Great improvements in transportation within the industrial mode mean that even bulky, heavy goods—e.g. timber or rocks, can be transported with ease over large distances (Ayers, 1978). 'The windmill gives you the feudal lord, the steam engine the capitalist', says Marx when distinguishing between different modes of production. Distinguishing between different modes of resource use according to their technological infrastructure, we may likewise say: 'the axe and bullock cart give you the peasant mode, the chain-saw and locomotive the industrial mode'.

These abilities *vis-à-vis* materials processing, storage and transport have revolutionized flows of resources, these having now become truly global. Any material, be it animal, vegetable, or mineral, natural or man-made, can now be rapidly transported anywhere, and whenever desired. With this a significant fraction of humanity, albeit still very much a minority, has come to consume vast quantities of a very wide range of resources (IIED and WRI 1987). Consumers in the high centres of industrial civilization can draw upon the natural resources of

most parts of the globe, taking for granted the continued supply of teakwood from India, ivory from Africa, and mink from the Arctic. This elite is now scarcely affected by the natural variations within resource availability in space and time; it has developed a lifestyle which could truthfully be called global. In the process it has wiped out myriad locally-adapted lifestyles in different parts of the world.

Economy: Over the last three centuries, industrial societies have steadily expanded their resource base. This has been achieved by a growing knowledge about the working of nature—through the hypothetico-deductive method of modern science and by the links established between scientific discovery and practical application—in order to tap additional sources of energy, process materials, and transport goods faster, more economically, and over ever-longer distances. This process of the intensification of resource use has led to the continual over-use and exhaustion of many resources. The typical response to such exhaustion has been to find a substitute, though initially, when the resource being substituted was abundant, the use of the substitute would have involved more effort in terms of energy, material, and labour. In the classic case, when wood became scarce in Europe, coal came to substitute for wood charcoal in the manufacture of iron, while locomotives were invented to replace horse-drawn carts even as there developed serious scarcities of fodder for horses. Industrial societies have gone on to consume resources at an accelerating rate, exhausting them one by one in sequence, going from the economically desirable resource of the moment to less and less desirable resources as the more desirable resources become exhausted. And, alongside, technological innovations now continually extract better and better results from what were once the less desired resources (Wilkinson 1988).

The expansion of the resource base in industrial societies has rested upon access to land and natural resources which were earlier controlled by gatherer and peasant societies. Reeling under an energy and transport crisis brought about by the

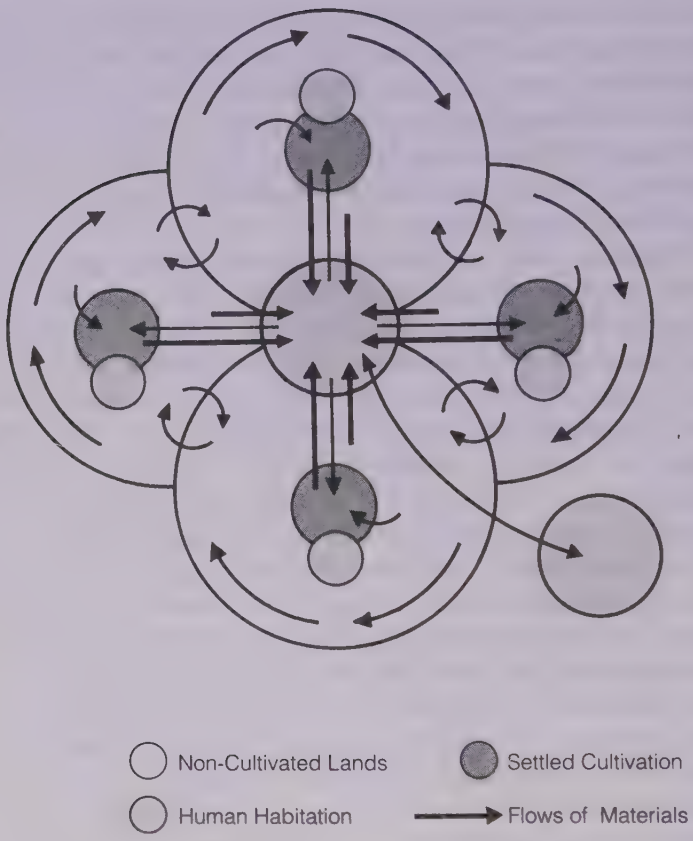


Figure 1.3
Material flows characterizing modern Indian society. Such societies not only tap surplus agricultural production, but also a great deal of the produce of non-cultivated lands to meet the requirements of the urban-industrial sector. Thereby the material cycles become totally open, with large outflows from rural hinterlands. These are partially compensated for by the organization of flows of materials such as fertilizers from the urban-industrial sector to the cultivated lands. The large central human habitiation represents an Indian city such as Bombay; the separated habitation in the upper right hand corner, the industrialized countries. The thickness of an arrow indicates the intensity of the flow.

exhaustion of their forests, European colonists laid claim to vast terrains the world over. Wherever the 'portmanteau biota' of these Europeans—wheat and cattle, their weeds and diseases—could establish itself comfortably—as in North America and Argentina, South Africa, Australia and New Zealand—it created neo-Europes (Crosby 1986). Where the ecological setting did not permit such a takeover, as in the older civilizations of the Middle East and Asia, or in the humid tropical forests of the Amazon, the Congo and Malaysia, these Europeans nevertheless established a firm hold on the resources of these regions and organized outflows of what they most desired to their own lands and people. Thus, India became an exporter of teak, cotton, jute, tea, indigo and precious metals; Burma of rice and teak; the West Indies of sugar; Brazil of rubber and coffee; and so on. Although colonialism has formally ended, this process vigorously continues, and India now exports prawns and trained manpower along with tea, rather than teak or cotton; and Brazil exports beef and forest produce instead of rubber.

These flows are highly asymmetric, with industrial societies receiving large volumes of unprocessed resources at low prices and exporting small volumes of processed resources at much higher prices. Simultaneously, this process also implies the production in industrial societies of high volumes of wastes; these are then sought to be disposed of elsewhere, either in the global commons of the oceans and the atmosphere, or through their sale to Third World societies plagued by foreign debt.

Social organization: The greatly enhanced scale of resource flows in industrial society goes together with a substantial increase in the number of humans involved in this network. Face-to-face contact is obviously impossible among such large numbers, and they tend to interact through the medium of formally codified transactions. Within such large social groups there is, of course, a rather elaborate division of labour. In modern industrial societies this division largely hinges upon skills in processing, transporting and exchanging materials and information. Smaller social groups based on such division of

labour—for instance, car mechanics or teachers—and corporate groups banded together to carry out a task—for instance, car manufacture or schoolteaching—become important. Among these exist relations involving face-to-face communication equivalent to those in hunter-gatherer bands or peasant villages. In these societies, too, men have tended to monopolize the more skilled and prestigious jobs, leaving the bulk of the unskilled and tedious work to women. Such discrimination has persisted despite the rhetoric of sexual equality and the development of technologies which tend to discount the advantages of greater physical strength.

Whereas kinship, locality and region define the forms of association in the pre-industrial world, in industrial societies the impersonal criteria of structural location *vis-à-vis* the means of production define the ways in which individuals come together for collective action. These corporate groups, based on the division of labour, are extremely fluid, the membership being in great flux. Again, the continual expansion of the resource base, both in terms of new extractive techniques and new territories to draw upon, has enhanced the fluidity of social groupings. There is therefore a much greater stress on pursuit of individual interest in these greatly atomized societies (Hawley 1986).

The rise of individualism is also accompanied by a tremendous expansion in the role of the state in regulating individual transactions. In most spheres of social life, the personalized and flexible systems of customary law—typical of agricultural societies—are replaced by impersonal and relatively rigid systems of codified law. The forest codes of modern society are amazingly detailed, often running into several hundred sections and subsections (cf. Guha, R. 1983; Merriman 1975; Linebaugh 1976). While safeguarding private property over land and in the workplace, and taking over the ownership of what was hitherto common property, the modern industrial state completely delegitimizes *community*-based systems of access and control.

Forest management provides a good illustration of how the industrial mode admits only the polarity of individual and state control. In the process of industrialization in Western Europe and the United States, an early phase of anarchic capitalist exploitation was followed by the assertion of state control, wherein the government stepped in to assume responsibility for forest protection and production (Hays 1958; Heske 1937). However, when the situation stabilizes with respect to the supply of raw material for industry, the state may withdraw control over certain forest areas by handing over 'captive plantations' to private industry. Simultaneously, the expansion of the market may encourage the widespread planting of commercially valued trees by individual farmers. Thus, forests in the industrial mode are primarily *state*-owned, supplemented in capitalist societies by a small but often growing *private* sector. The victim in this process is, inevitably, village-based systems of *community* forests and pasture management: the lands in these systems are either sequestered by the state or parcelled out among individuals.

Ideology: Developed in a mileu of r-strategists, the ideological underpinnings of industrial society involve the total rejection of the gatherer view of man as part of a community of beings, or even of the agriculturist view of man as a steward of nature. Instead it is emphatically asserted that man is separate from nature, with every right to exploit natural resources to further his own well being. Nature is now desacralized. What has come to be venerated instead is the marketplace; the market is supposed to rationally allocate the use of resources so efficiently that all individuals are as well off as they could possibly be. Among socialists this veneration is transferred to a Central Plan: it is regarded by them as a more efficient allocator than the market.

While the ideology of conquest over nature as well as modern lifestyles have resulted in a radical alteration of the landscape of the globe, industrial societies have, to be fair, made systematic attempts to safeguard their own environments. Such

attempts were triggered off by deforestation and the consequent landslides in the Swiss Alps in the 1860s, and further strengthened when the western frontier of the United States was officially declared closed by the 1890 census. Since then Western conservationists have gradually brought back part of the forest cover of Europe and North America, set in motion an effective programme of soil conservation, and set apart wilderness areas (cf. Hays 1958 and 1987; Nash 1982). The problems of chemical pollution which surfaced after the Second World War are also being tackled—with the successful cleaning of the Thames, for instance. But pollution proceeds apace with such efforts; new environmental issues come up, such as acid rain and global warming. These require global solutions, so that it now appears as if industrial societies can no longer restrict environmental degradation to areas outside their own borders (WCED 1987).

Despite the increasing scale of environmental problems, the ideology of industrialism even today rejects as invalid any concern with physical 'limits to growth', trusting technical innovation to take care of problems as they arise. It puts its faith in the efficacy of the marketplace (or of the Central Plan) to stimulate the discovery of substitutes, and to enhance the efficiency of resource use (Beckerman 1972; Laptev 1977). A variety of scientifically-based prescriptions on sustainable use of soils, forests, fisheries and other renewable resources, generated over the last century, have been interpreted as examples of efficient resource use, and one of the aims of this book is to examine whether this is actually so.

Scientific prescriptions for resource use: Just as religion and custom legitimized patterns of resource use in pre-industrial societies, in the industrial mode science provides the organizing principle for human interactions with nature. However, scientific prescriptions for the preservation, the sustainable use and the deliberate destruction of different species and habitats parallel, in many ways, the pre-scientific prescriptions typical of earlier modes. For example:

• Just as there were quantitative restrictions on the amount of fuelwood extracted by a family from a community wood-lot, or on the number of animals to be grazed on a fodder reserve, there are now quantitative prescriptions on the amount of timber to be removed during the course of selection felling from a designated patch of forest, or on the number of deer that may be shot by a licence-holder in a given year.

• Just as New Guinea hunter-gatherers stop hunting a particular species—say the Bird of Paradise, when its population density falls below a certain level—the International Whaling Commission has banned the hunting of certain species of whales whose populations have been severely depleted.

• Just as fishing from certain coastal areas is traditionally stopped if the yields become very low, there have been prescriptions to 'rest' certain forest areas, for instance in India after the excessive harvests of the Second World War.

• Just as there is a traditional ban on hunting in certain seasons, there are closed seasons for mechanized fishing, or for timber extraction.

• Just as certain life-history stages are traditionally given protection—for instance, breeding birds at heronries in many villages in peninsular India—the younger, growing stock in a forest is supposed to be spared all extraction pressure.

• Just as certain methods of harvest, e.g. the poisoning of streams, may be traditionally forbidden, there are regulations forbidding fishing with nets of excessively small mesh size.

• Just as certain habitat patches—for instance, sacred groves or sacred ponds—may be fully protected from harvest, there are prescriptions for keeping certain forest areas totally free from human interference, either for watershed conservation or for the maintenance of biological diversity.

• Just as certain species are totally exempt from being hunted or felled, e.g. monkeys or *Ficus* trees in Indian villages, complete protection may be extended to certain endangered species, such as the California condor or the whooping crane.

In parallel, of course, there exist a whole range of supposed-

ly scientific prescriptions for deliberately destroying certain resources. Thus, just as the entire Khandava forest near present-day New Delhi was supposedly burnt in the time of the Mahabharata (cf. Chapter 2) as an offering to the fire god, massive forest areas in the Narmada valley in Central India are now being destroyed to create reservoirs for hydro-electric power generation and irrigation. And one cannot forget that the large-scale defoliation of forests during the Vietnam war was supported by considerable inputs from the American scientific commuity.

In this manner, scientific prescriptions in the industrial mode closely parallel traditional, 'pre-scientific' prescriptions based on a large but informal base of knowledge and on simple rules of thumb. In part, modern prescriptions are based on a more detailed knowledge of the behaviour of the system, and on a more explicit definition of what is sought to be achieved. With many tree or fish populations, this aim is defined as achieving maximum sustainable yield, i.e. a yield that will not decline in the long run.

However, deciding upon a harvesting regime that would lead to maximum sustainable yields is beset with manifold difficulties (Clark 1985; Beddington and May 1982). First, every plant or animal population is involved in a number of interactions with other members of its community. Thus, harvesting a deer population may, through a change in the composition of the herb layer, affect another resource such as wild cattle. Or harvesting a plant species may result in a change in the availability of honey. In fact, some years ago, biologists warned the fishing industry on the east coast of Canada that scallop fisheries were being over-exploited and were likely to crash. It so happened that just after this there was an outbreak of an epidemic disease among sea urchin populations which prey heavily upon the scallop. With predation pressures reduced, scallop populations increased rather than collapsed, as had been feared by fishery biologists (cf. Berkes 1989). Similarly, wild-life biologists concerned with enhancing the carrying

capacity of wetlands within the Bharatpur bird sanctuary in north India recommended the banning of buffalo-grazing in the sanctuary. Following the imposition of this ban it was found that, in the absence of grazing, a grass, *Paspalum*, grew so rapidly that it choked the wetland and reduced its carrying capacity for water fowl (Vijayan 1987).

Because of such complex community interactions, the simultaneous harvesting of more than one resource population from a specific locality can have a variety of unexpected consequences. Theoretical investigations suggest that in some cases the natural tendency of a population to fluctuate might be exaggerated by the simultaneous harvesting of another species, and expose it to the danger of extinction (Pimm 1986; May 1984).

But, aside from such complications which arise out of community interactions, it is difficult to understand the behaviour of harvested populations and to assess whether they run the risk of being over-harvested and becoming extinct. For example, the Elk populations of British Columbia are being simultaneously harvested—for subsistence by Amerindians and for sport by European settlers. Wildlife biologists primarily interested in deciding on regulations for sport hunting contend that the Elk populations are declining on account of subsistence hunting. On the other hand anthropologists looking at the same population data suggest that there is no good evidence of a decline in Elk populations (Freeman 1989). It is even more difficult to assess the fate of fish or whale populations for which the only available information is based on harvests themselves. The level of such harvests is a complex function of the number, age and sex composition of the prey population, its distribution in space and time, and the extent and distribution of the harvesting effort. The parameters which characterize the prey population also depend on the extent of the exploitation to which it has been subjected in the past. It is difficult to predict the responses of such populations to different levels of harvests even on the basis of sophisticated mathematical models because our knowledge of many parameters in the model remains

limited. Indeed the International Whaling Commission finds it very difficult to arrive at a consensus on what is actually happening to many whale populations (May 1984).

Given all these difficulties, even today successful prescriptions of how to obtain maximum sustainable yields are possible only in very limited contexts. They work only when community interactions, as well as the population structure, are drastically simplified. This is the case with single-species even-aged populations of forest-tree species such as pines, planted for harvest in temperate latitudes. Here all other plant, insect, and microbial species that might enter the forest naturally are sought to be eliminated. Intra-specific competition is regulated by permiting reproduction only through artificial regeneration, by controlling inter-plant distance, and by harvesting the individual plants which are being suppressed through intra-specific competition. It would not be incorrect to say that scientific prescriptions on maximum sustainable yields have largely failed to work, except in such limited contexts. Even in these contexts, the suspicion is that a gradual exhaustion of soil nutrients is now leading to the decline of forests which have been long invoked as exemplars of sustained-yield management. We shall later see, in much greater detail (Chapter 6), how modern attempts to apply such concepts to the tropical forests in India have led to serious over-harvests (see also FAO 1984).

The prescriptions on giving total protection to certain species and localities, such as those embedded in the US Endangered Species Act or CITES (Convention on International Trade in Endangered Species), or to certain localities—as in programmes setting apart national parks or biosphere reserves—also have rather uncertain scientific foundations. This is inevitable, given our very limited understanding of the extent and distribution of biological diversity, and of the ecosystem processes which govern the survival of the species which make up different biological communities. Heated debates on whether to aim for a few large, several small, or even plentifully patchy nature reserves, as well as the debates on minimum

viable population size, show that scientists are still far from finding definitive answers on the appropriate ways of conserving natural populations of living organisms (Soule 1986). Indeed one could argue that scientific prescriptions in industrial societies show little evidence of progress over the simple rule-of-thumb prescriptions for sustainable resource use and the conservation of diversity which characterized gatherer and peasant societies. Equally, the legal and codified procedures which are supposed to ensure the enforcement of scientific prescriptions work little better than earlier procedures based on religion or social convention.

Ecological impact: Industrial societies, unlike gatherer or agricultural societies, are no longer directly dependent on the natural resources of their immediate vicinity. At first glance this may suggest that they could effectively conserve their resource base, whereas societies that follow more primitive modes of resource use would perforce suffer a much more adverse impact. This seems borne out by the fact that Japan, the most industrialized Asian country, has the best-preserved forest cover, while the forests of countries like Malaysia and Indonesia —which have large populations dependent on primitive agriculture—are being devastated. But a second look shows that Japan maintains its forest cover, in spite of its enormous per capita consumption of timber, only by shifting the pressure on to Malaysia and Indonesia, precisely the countries which suffer devastation. The Japanese also exploit their coastal fisheries very conservatively, while refusing to give up whaling in international waters. This discrepancy between prudence at home and profligacy abroad also characterizes the behaviour of other industrially advanced countries such as the USA and the (former) USSR.

If one looks at the overall picture, therefore, it is obvious that the enormous resource demands and waste production of industrial nations, and of the industrial segments in less industrialized nations, have the most profound impact on the world's environment (cf. Bahro 1984). This impact includes

radical modifications of the landscape, as with the laying of the Alaskan pipeline, or cattle ranching in Amazonian forests; a gradual depletion of forests, as with acid rain in Europe or over-extraction in Malaysia and Kalimantan; drastic reductions in, or extinctions of, populations, as with the passenger pigeon and the bison in the last century, or the African elephant and myriad other species in Amazonia in this century; a wholesale poisoning of the biosphere, as with wide-spectrum pesticides and nuclear waste; a modification of bio-geochemical cycles, as with the increased production of carbon dioxide; and perhaps a long-term adverse modification of the climate as well (cf. Richards 1986).

In assessing the ecological impact of different modes, one is struck by two paradoxes, which we illustrate here with respect to forest use. Spatially, hunter-gatherers live *in* the forest, agriculturists live adjacent to but within *striking distance* of the forest, and urban-industrial men live *away* from the forest. Paradoxically, the more the spatial separation from the forest, the greater the impact on its ecology, and the further removed the actors from the consequences of this impact! The same conditions operate with regard to other resources, such as water.

Second, the faster the development of formal, scientific knowledge about the composition and functioning of forest types, the faster the rate of deforestation. One important reason here is undeniably higher levels of economic activity, but another though less obvious factor is the *idiom* of resource use itself. While enormously enlarging our knowledge about specific physical and chemical processes, modern science has not always displayed the same understanding of the ecological consequences of human interventions which follow the development of scientific knowledge. However, the *belief* that science provides an infallible guide has nonetheless encouraged major interventions in natural ecosystems, and these have had unanticipated and usually unfortunate consequences. The history of both fisheries and forest management are replete with illustra-

tions of the failure of sustained-yield methods to forestall ecological collapse (McEvoy 1988; FAO 1984). While there no longer exist social constraints on levels of intervention, scientists are also inadequately informed on the ecological processes in operation. Ironically, therefore, religion and custom as ideologies of resource use are perhaps better adapted to deal with a situation of imperfect knowledge than a supposedly 'scientific' resource management.

Conflict Between and Within Modes

Inter-modal conflict: As one mode of resource use comes into contact with another mode organized on very different social and ecological principles, we expect the occurrence of substantial social strife. In fact, the clash of two modes has invariably resulted in massive bursts of violent and sometimes genocidal conflict. One of the best documented of such conflicts is the clash between the indigenous hunter-gatherer/shifting-cultivator populations of the New World on the one hand, and the advance guard of European colonists practising an altogether different system of agriculture on the other. Historians of the victorious white race have in recent years depicted, with marked sensitivity, the ecological bases of a conflict that resulted in the extermination of the bulk of the native population and the traumatization of the segment which escaped annihilation (Cronon 1983; Crosby 1986). Yet, as Chapter 2 of this book argues, such episodes are not exactly foreign to the histories of the Old World. For the brutal conflict between the American Indians and the English colonists was anticipated several millenia before—in the conquest of indigenous hunter-gatherer populations within India by invading agriculturists, a clash vividly captured in the sacred text of the conquerors, the Hindu epic Mahabharata.

The environmental and social costs of the encounter between agrarian and industrial modes in Europe have been ably chronicled by a long line of distinguished historians. The rise

of industrial capitalism radically altered relations not merely in land and the workplace, but also around the utilization of nature. The conflicts over the enclosure of what was previously common land, and the assertion of state control over forests, while perhaps not as brutal as the clash between hunter-gatherers and the neolithic vanguard, also exacted a heavy human cost (cf. Agulhon 1982). As one American forester, documenting the enclosure of the forests by the state and lords in Germany, laconically put it:

> All these changes from the original communal property conditions did not, of course, take place without friction, the opposition often taking place in peasants' revolts; hundreds of thousands of these being killed in their attempts to preserve their commons, forests and waters free to all, to re-establish their liberty to hunt, fish and cut wood, and to abolish titles, serfdom and duties (Fernow 1907, p. 34).

Even where there was no open revolt, peasants resorted to poaching and the theft of forest produce. These forest 'crimes' were very widespread. In Prussia in 1850, 265,000 wood thefts were reported, as against only 35,000 cases of ordinary theft. In this class struggle over nature, between the peasants on one side and landlords and the state on the other, the wood thieves were 'defending their entire economic system—the family economy which was based on collective usage rights' (Mooser 1986). Indeed until well into the present century, European foresters were unable to fully extinguish the customary rights of user granted as a consequence of the peasant opposition to state forest management (cf. Heske 1937). In the American South too, common rights of hunting and grazing on non-cultivated land withstood landlord and state pressures until the early decades of this century (Hahn 1982).

The European experience is directly comparable to the Indian (detailed in chapters 5 and 7), where the encounter between peasant and industrial modes, mediated partly by colonialism, has also greatly intensified social conflicts over

forest resources. In the colonial societies of South East Asia and Africa as well, the takeover of forest land for strategic and commercial purposes fuelled bitter conflicts between the state and the peasantry (Scott 1976; Grove 1990). Here we must note that colonial methods of ecological control also intensified conflicts within the agrarian sector. Two examples come to mind. One is the clash over forests, lands and water in colonial and semi-colonial societies, between large cash-crop plantations and the traditional peasantry (Womack 1969; Pandian 1985). The other is the restrictions placed on subsistence hunting-gathering and grazing by the setting up of game reserves by white colonists (cf. Kjekshus 1977).

While gatherer–peasant and peasant–industrial conflicts are both vividly represented in myth and history, a third inter-modal conflict dominated medieval Europe and medieval Asia, namely between the peasant and pastoral modes. Despite the many instances of nomad–peasant symbiosis, for example the pasturing of livestock on fallow fields in return for manure, this relationship has historically been fraught with friction. Medieval sources in the Near East talk of nomads driving their livestock to graze on *sown* fields—a practice beneficial to the pastoralists but disastrous for cultivators. Conflicts have also arisen when agriculturists shift to crops (e.g. cotton) which do not provide stubble during the dry seasons. In this conflict between two modes of resource use that have, for the most part, overlapped in time and space, the nomads have on occasion greatly expanded their resource base—as with the Mongols of Central Asia in the medieval period—while at other times their own niche has steadily shrunk—as with the reservation of forests and the expansion of irrigated agriculture in modern India (Khazanov 1984; Gadgil and Malhotra 1982).

Our final example of inter-modal conflict comes from the contemporary West. In recent years a distinctively new form of ecological conflict has surfaced, between votaries of the industrial mode of resource use and a mode struggling to be born. Thus, while scientific foresters and industrial users continue to

look upon the forest primarily as a resource to be harvested, in the perspective of the Western environmental movement forests are to be preserved as a haven from the workaday world and as a reservoir of biological diversity. Although this conflict has by no means been as violent as those between earlier modes of resource use, the ideological differences between the industrial mode and what environmentalists call the 'post-industrial' mode can hardly be minimized (Nash 1982; Hays 1987).

Two aspects of inter-modal conflicts need to be highlighted. First, apart from massive bursts of social conflict, the encounter between the different modes also signals a spurt in rates of ecological destruction. The deforestation of the Indo-Gangetic plain in the first and second millennium BC, the draining of the fens in eighteenth-century Europe, and the destruction of forest cover in colonial India—all these testify to the enormous environmental costs associated with the advent of a new mode of resource use. (Of course the emergence of a 'post-industrial' mode, if it occurs, may reverse this trend.) As one mode wins out, there is a slow but perceptible diminution in the levels of social conflict and ecological disturbance.

Second, while the conflict between different modes of resource use is, at the most elemental level, a struggle for control over productive resources, it is invariably accompanied by an ideological debate legitimizing the claims of the various modes. Agrarian societies have typically justified their takeover of the lands and resources of gatherer societies in terms of the latters' low productivity and 'wasteful' use of nature, a distinction used by colonists in America for their conquest of Indian territories (Cronon 1983; Prucha 1985). Likewise, votaries of the industrial mode have used the rhetoric of scientific conservation when legitimizing their claims. The scientific model of natural resource management, it is claimed, is a distinctively modern innovation, and inherently superior to the idioms of religion and custom which legitimized human interactions with nature in earlier epochs. And, as the widespread disaffection with

excesses within the industrial mode finds expression in modern environmental movements, it is no accident that religion and custom are upheld as being, after all is said and done, more prudent in their use of nature than modern 'scientific' methods.

Intra-modal Conflict

The ebb and flow of social conflict *within* different modes of resource use is perhaps not as visible in history and myth as the violent struggles *between* modes. Yet such conflict is by no means absent. Most of the primate relatives of human beings are engaged in struggles to hold on to and expand group territories, and it is very likely that hominids were similarly engaged ever since their origins. There is little fossil evidence of violent death in the period before 30,000 b.p., when modern *Homo sapiens*, with its capacity for symbolic communication, finally came into its own. Since that time, conflicts relating to territorial control have undoubtedly been a feature of most hunter-gatherer societies, as documented by anthropologists in such far-flung areas as New Guinea, New Zealand, and the Amazon (cf. Rappaport 1984; Vayda 1974).

Such intra-modal conflicts become more complex as one passess on to more advanced modes of resource use. They become especially acute when the ideal-typical characteristics of a mode, as outlined earlier in this chapter, are perceived as being distorted to subserve the ends of a particular social group. When feudal lords fail to honour the customary codes of the moral economy, for example, or when the workings of forest law and science are seen to be class-specific rather than class-neutral, the ideological basis of the mode of resource use begins to crumble. At such times an idiom of conflict, rather than collaboration, characterizes intra-modal relations.

Coming to the peasant mode, in many periods conflicts between the peasantry and overlords over natural resources has been endemic. In England, for example, conflicts over rights in forests were especially acute in the thirteenth and fourteenth

centuries, a period of rising demographic pressure and the expansion of arable land. There were growing numbers of prosecutions for timber theft, with peasants also occasionally invading woodland enclosed by lords and abbots (Faith 1984; Birrel 1987). Conflicts over forest and pasture were an important element in one of the greatest ever anti-feudal revolts, the German Peasant War of 1525 (cf. Engels 1956). And in France from the sixteenth to the eighteenth centuries, peasants repeatedly rose in revolt against attempts by landlords to usurp forests, swamps and grazing grounds earlier held in common (Ladurie 1980; Bloch 1978). Finally, resistance to the enclosure of community pastures by plantations was widespread in early-twentieth-century Mexico (Lewis 1964; Womack 1969).

Another form of intra-modal conflict, characteristic of mature feudalism in Europe, related to peasant rights over pasturage and timber in forests which were reserved exclusively for hunting by the nobility (Thompson 1975; Hay *et al.* 1975). While in normal times peasants were unable to challenge this monopoly (though they continued to breach it on the sly), when the state was vulnerable they quickly and forcefully asserted their rights. Thus, in the rural revolts that accompanied the French Revolution, groups of peasants broke into the hunting preserves of the nobility and 'determinedly hunted the game' (Lefebvre 1982, p. 44 and *passim*). Similar invasions of forests controlled by the nobility were also reported in the wave of peasant strikes during the Russian Revolution of 1905 (Shanin 1986).

With the shift from the peasant to the industrial mode, conflicts within the peasant mode intensify as one class within agrarian society is quicker to adapt to the socio-ecological orientation of the coming mode (e.g. landlords, in the case of the enclosure movement). But the internal conflicts characteristic of the industrial mode—particularly its capitalist variant—are quite different. These are, first, the continuing struggles of individual capitalists with each other, and against the state, for formal proprietorial control of non-cultivated areas,

as well as the terms of disposal of living resources. Second, the industrialization of forest usage itself creates a new class of workers whose interests are not always in harmony with those of capitalists in the timber harvesting and processing sectors (Vail 1987).

These forms of intra-modal conflict, and the ways in which they are resolved, shed light on the interlinkages between modes of production (defined in the Marxian sense) and their corresponding modes of resource use (as defined here). In the natural economy of hunter-gatherers, of course, the mode of production is simultaneously the mode of resource use. In the peasant and industrial modes the links are more complex. In the former case, even where relations around land are accepted (however grudgingly) as asymmetrical, peasants insist that they have full access to the 'free gifts' of nature. And when enclosure-minded landlords and hunting monarchs violate this tacit agreement by restricting the exercise of common rights, peasants resist and the levels of conflict escalate. For the *stable* functioning of the mode of production, therefore, there must exist some discrepancy between the rights of the overlord to land (and a portion of its produce), and to living resources, respectively.

The stable functioning of the industrial-capitalist mode likewise requires a discrepancy between property relations in the field/factory, and in the forest. While private ownership predominates in the former case, forests are, to a much greater extent, owned and controlled by the state. However, the underlying rationale of government intervention is precisely to safeguard the stability of the industrial mode, by taking the long-term view available only to the state, thereby harmonizing conflicts between individual capitalists.

TABLE 1.1

FEATURES OF TECHNOLOGY AND ECONOMY IN A SOCIETY PREDOMINANTLY FOLLOWING ONE OF THE FOUR MAJOR MODES OF RESOURCE USE

	Gathering (including shifting cultivation)	*Nomadic pastoralism*	*Settled cultivation*	*Industry (including fossil-fuel-based agriculture)*
Energy resources used	Human muscle power, fuelwood	Human and animal muscle power, fuelwood	Human and animal muscle power, fuelwood; coal, water power to some extent	Fossil fuels, hydro-electricity, nuclear power; fuelwood, human and animal muscle power much less important
Material resources utilized	Stone	Plant and animal material	Stone, plant and animal material, some uses of metals	Extensive use of metals and synthetic materials
Abilities to store resources	Very rudimentary	Domestic animals serve as meat supply on hooves	Grain and domestic animals make long-term storage of food possible	Even highly perishable materials like fleshy fruit and meat can be stored over long periods
Abilities to transport resources	Very rudimentary	Domestic animals like horses make long distance transport possible	Domestic animals make long distance transport possible	Fossil-fuel based vehicles render transport over great distances easy

	Gathering (including shifting cultivation)	*Nomadic pastoralism*	*Settled cultivation*	*Industry (including fossil-fuel-based agriculture)*
Abilities to transform resources	Very rudimentary	Rudimentary	Low, including metal making, weaving	Very extensive
Spatial scale of resource catchments	Small, mostly of order of a few hundred or thousand km^2	Could be quite extensive	Moderate	Global
Quantities of resources consumed	Very moderate	Moderate for most	Moderate for most, a small elite may consume large quantities of commodities	Large numbers consume enormous quantities

TABLE 1.2

FEATURES OF SOCIAL ORGANIZATION AND IDEOLOGY IN A SOCIETY PREDOMINANTLY FOLLOWING ONE OF THE FOUR MAJOR MODES OF RESOURCE USE

	Gathering (including shifting cultivation)	*Nomadic pastoralism*	*Settled cultivation*	*Industry (including fossil-fuel-based agriculture)*
Size of social groups	Small, a few thousand people	Moderate, several thousand people	Moderate, several thousand people	Very large, in hundreds of thousands
Extent of kinship within social groups	Very strong	Strong	Strong, but interactions with non-kin increasing	Very weak
Extent of attachment of social groups to particular localities	Often strong	Weak	Often strong	Very weak
Division of labour	Rudimentary	Sex-age based	Rudimentary sex-age based	Considerable, based on specialized skills and knowledge
Role of division of labour in formation of social groups	Very weak	Very weak	Moderate to strong	Strong

	Gathering (including shifting cultivation)	*Nomadic pastoralism*	*Settled cultivation*	*Industry (including fossil-fuel-based agriculture)*
Idiom of social transactions	Totally informal, based on face-to-face contact	Informal, primarily based on face-to-face contacts	Social conventions and codified transactions	Codified transactions with legal sanctions very important
Within group differentiation of access of resources	Weak	Weak	Considerable	Very extensive
Mechanisms governing access to resources	Community-based decisions	Community-based decisions	Private control of farmland, community and state control of non-cultivated lands	Private, state and corporate ownerships predominant; community ownership delegitimized
Perception of working of nature	Nature viewed as autonomous, capricious	Nature largely seen as capricious	Nature viewed as partially law-bound, controllable	Nature viewed largely as lawful
Idiom of man-nature relationship	Man part of community of beings	Man potentially conqueror of nature	Man a steward of nature	Man above and apart from nature, fully capable of controlling it

Recapitulation

The salient features of the four modes of resource use are reproduced in Tables 1.1 to 1.4. To recapitulate, as human societies move temporally from hunting and gathering through pastoralism, agriculture, and finally into industrialization, five distinct but closely interrelated processes occur. First, there is an increasing intensity of resource use and exploitation. Second, there is a secular increase in the level of resource flows across different geographical regions and across different levels of any economic/political system. Third, there is an integration of larger and larger areas into the domain of any given political/economic system. Fourth, there is, at the global level, a secular increase in population densities and in the extent of stratification and inequality with respect to the access, control and use of different natural resources. Finally, there is an intensification of rates of ecological change and ecological disturbance.

Of our four modes of resource use, in the Indian context nomadic pastoralism is best treated not as a separate mode but as being integrated with the peasant mode of resource use, within whose ecological zone it occupied a special niche. In Parts II and III we shall move from the ideal-typical to the historical, narrating the rise, decline and fall of different modes of resource use in the Indian subcontient.

TABLE 1.3

THE NATURE OF THE ECOLOGICAL IMPACT IN SOCIETIES PREDOMINANTLY FOLLOWING ONE OF THE FOUR MAJOR MODES OF RESOURCE USE

	Gathering (including shifting cultivation)	*Nomadic pastoralism*	*Settled cultivation*	*Industry (including fossil-fuel-based agriculture)*
Land transformation	Little, some regression of patches of forest to successional stages or grassland	Some extension of grasslands, deserts	Forests, grasslands extensively converted to fields	Large-scale deforestation, desertification, built-up habitats
Habitat diversity	Enhanced	Somewhat reduced	Reduced	Substantially reduced
Biodiversity	Little affected	Some effect	Moderate effect	Considerable impact
Resource populations	Occasionally overharvested	May overgraze some grasslands and overharvest some prey populations	May overgraze some grasslands and overharvest some prey populations	Many resource populations overharvested
Substances poisonous to life	Nil	Nil	Nil	A large range of synthetic chemicals
Modification of biogeochemical cycles	Very little	Very little	Little	Substantial
Modification of climate	Highly unlikely	Highly unlikely	Unlikely	Quite likely

TABLE 1.4

FEATURES OF SOCIAL ORGANIZATION, IDEOLOGY AND ECOLOGICAL IMPACT IN SOCIETIES IN RELATION TO CHANGES IN THEIR RESOURCE BASE

	In equilibrium with resource base	*Resource base expanding*	*Resource base shrinking*
Fluidity of social groups	Low	High	Often considerable
Extent to which group interests prevail over individual interests	Group interests quite significant	Individual interests more important	Group interests may crumble
Perception of man-nature relationship	Man as steward of nature	Man as conqueror of nature	Man helpless
Extent to which sustainable resource use prevails	Quite often	Rarely	Rarely
Level of ecological impact	Low to moderate	High	High

Appendix to Chapter I

Note on Population

In our analysis of modes of resource use, we have classified these modes according to technology, pattern of resource flows, social structure, dominant ideologies, and systems of conservation. In this appendix we deal briefly with another dimension which has an important bearing on ecological history—namely the density of human population. The reflections which follow are prompted by the question—under what conditions will human populations regulate rates of reproduction in individual or group interests?

In utilizing the resources of their environment at very low intensities, gatherers also maintain low population densities. As suggested above, in stable environments they are likely to maintain fixed territories, with high levels of conflict with neighbouring groups over territorial control. Under these conditions their interests lie in maintaining the population density at a level which ensures that the territorial group does not have to face occasional resource shortages—shortages that may weaken it in a conflict with neighbouring groups. It is possible that the variety of population-control mechanisms noted among gatherers may serve the function of maintaining their population densities in the group's interest. Such mechanisms may, of course, break down when gatherers lose control over their resource base—upon coming into contact with people practising more advanced modes of resource use.

Coming next to pastorals, their population densities tend to be low because they inhabit the more arid and unproductive regions. Very likely, these levels are maintained by natural checks; it is unlikely that the inhabitants of such variable environments, who have no fixed territories, would have evolved cultural practices to deliberately maintain their populations at low levels.

Characterized by a substantial intensification of resource

use, agricultural societies maintain population densities far higher than those of either gatherer or nomadic pastoral societies. However, most members make rather low-level demands on the resource base, though members of the non-agricultural elite may have high levels of demand for many non-essential commodities. Given these low levels of resource demand, and the possibilities of gradually improving yields from cultivated lands through technical change, agricultural communities are likely to be characterized by slow population growth. Further, the larger states of which they are a part generally ensure that there are no serious territorial conflicts within their limits—conflicts in which overpopulation may be a serious handicap. Yet the military strength of agricultural states may depend on their overall population; therefore the state apparatus would tend to encourage population growth. One therefore expects neither individual nor group-level pressures to deliberately check population growth in agricultural societies; *natural* checks and balances, such as diseases and disasters, may take a periodic and heavy toll of peasant populations.

The population history of the industrial mode is captured in the phrase 'demographic transition'. While in the initial phase of industrialization the population of European stock (both within Europe and in newly-colonized lands) expanded rapidly, over the last century it has grown much more slowly, stabilized, and in some places even declined. This transition seems related to the fact that the attempts of each individual in enhancing his/her resource consumption limits the quantity of resources available to raise offspring. Further, parents try to endow each offspring with a high level of ability to garner resources for itself. The need to invest in the *quality* of children again implies a severe limitation on the *quantity* of children produced. Industrial societies have thus generally stabilized their populations. At the same time, their *per capita* resource consumption remains high and on the increase.